

Supplementary Materials for “Long-Tailed Specific Emitter Identification: A Curriculum Learning Method with Multi-Domain Feature Fusion”

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I. INTRODUCTION

This supplementary material analyzes the consistency of samples generated by adaptive synthetic sampling (ADASYN) to verify their ability to preserve the class-discriminative fingerprints of automatic dependent surveillance–broadcast (ADS-B) emitters, as discussed in the main manuscript [1]. We conduct a 10-category experiment to validate the data augmentation strategy by comparing the feature-space distributions of real and synthetic samples. In addition, controlled replacement experiments are conducted to evaluate the tail-class augmentation method, curriculum learning strategy, and multi-domain feature fusion method.

II. CONSISTENCY ANALYSIS OF ADASYN-GENERATED SAMPLES

A. Experimental Settings

The experiment uses the ADS-B dataset [2] with 10 categories. For each class, only 200 real samples are used to train the model, while 50 real samples are used for validation and another 50 for testing. The training framework follows the proposed method, where four-domain feature representations are extracted by four ResNet-18 branches and fused through the weighted cyclic cross-attention module. In addition, 50 independent real samples per class are used as source samples to generate 50 synthetic samples via ADASYN [3]; these synthetic samples are not involved in model training. After training, the 50 real test samples and the 50 ADASYN-generated samples of each class are separately fed into the trained model, and their deep-feature distributions in the feature space are visualized for comparison.

B. Feature Visualization

Fig. 1 visualizes deep features of real and ADASYN-generated samples using t-SNE. Circles denote real test samples, crosses denote ADASYN-generated samples, and colors indicate categories. The figure shows clear clusters for each class, with synthetic samples lying within or near the

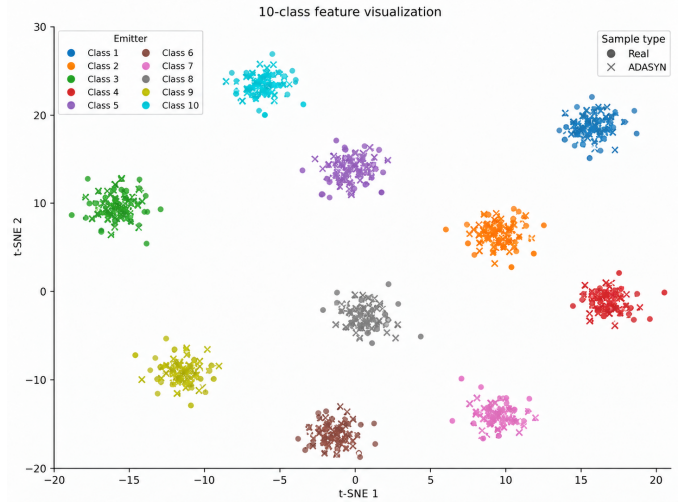


Fig. 1. t-SNE visualization of real and ADASYN-generated samples.

corresponding real-class regions rather than forming separate clusters. This experiment confirms that ADASYN preserves class-discriminative characteristics.

III. REPLACEMENT ANALYSIS OF THE PROPOSED COMPONENTS

Controlled replacement experiments are conducted on the 10-category long-tailed ADS-B dataset. In each experiment, only the target component is replaced, while the remaining network structure, data partitions, and training settings are kept unchanged.

A. Replacement of the Tail-Class Augmentation Method

We compare random oversampling, the synthetic minority oversampling technique (SMOTE) [4], and the ADASYN augmentation. Unlike random duplication and uniform SMOTE interpolation, ADASYN emphasizes difficult tail-class samples, resulting in better balanced recognition performance, as shown in Table I.

B. Replacement of the Curriculum Learning Strategy

We compare direct full-dataset training, reverse curriculum learning, and the proposed curriculum learning strategy. The proposed easy-to-hard strategy progressively introduces tail classes and their synthetic samples, thereby reducing early

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TABLE III
COMPARISON OF DIFFERENT MULTI-DOMAIN FEATURE FUSION METHODS ON THE 10-CATEGORY ADS-B DATASET

Metric	Feature concatenation	Average fusion	Standard cross-attention	Unweighted cyclic cross-attention	Weighted cyclic cross-attention
All (%)	90.41	89.76	91.22	91.83	94.06
Majority (%)	97.86	96.91	95.20	94.58	98.16
Minority (%)	78.63	80.27	85.60	87.46	91.09
Macro-F1 (%)	87.42	88.15	89.85	90.72	92.58

TABLE I
COMPARISON OF DIFFERENT TAIL-CLASS AUGMENTATION METHODS

Metric	Random oversampling	SMOTE-based augmentation	ADASYN
All (%)	84.35	86.94	94.06
Majority (%)	93.44	95.72	98.16
Minority (%)	75.86	76.18	91.09
Macro-F1 (%)	81.37	84.63	92.58

TABLE II
COMPARISON OF DIFFERENT TRAINING STRATEGIES

Metric	Direct full-dataset training	Reverse curriculum	Curriculum learning
All (%)	90.73	87.36	94.06
Majority (%)	94.81	90.84	98.16
Minority (%)	87.98	85.92	91.09
Macro-F1 (%)	89.10	86.47	92.58

REFERENCES

- [1] Z. Chai, J. Li, L. Yang, Z. Zhao, and L. Lv, "Long-tailed specific emitter identification: A curriculum learning method with multi-domain feature fusion," *IEEE Communications Letters*, 2026, submitted for publication.
- [2] Y. Tu *et al.*, "Large-scale real-world radio signal recognition with deep learning," *Chin. J. Aeronaut.*, vol. 35, no. 9, pp. 35–48, Sep. 2022.
- [3] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *Proc. IEEE Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2008, pp. 1322–1328.
- [4] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.

training instability and majority-class dominance, as shown in Table II.

C. Replacement of the Multi-Domain Feature Fusion Method

We compare feature concatenation, average fusion, standard cross-attention, unweighted cyclic cross-attention, and the proposed weighted cyclic cross-attention. Unlike feature concatenation and average fusion, which do not explicitly model cross-domain interactions, the proposed method exploits complementary information among different feature domains. Compared with standard cross-attention, the cyclic interaction design enables more comprehensive cross-domain information exchange. Moreover, compared with unweighted cyclic cross-attention, the learnable fusion weights can adaptively emphasize more discriminative domain information and suppress redundant features. As shown in Table III, the proposed method achieves better balanced recognition performance.

IV. CONCLUSION

The t-SNE results demonstrate that ADASYN-generated samples share similar feature distributions with real samples and retain class-related discriminative information. The replacement experiments further demonstrate that the tail-class augmentation method, curriculum learning strategy, and weighted cyclic cross-attention module effectively alleviate sample imbalance, insufficient minority-class representation, and classification bias in long-tailed SEI scenarios, respectively, verifying their effectiveness in addressing the long-tailed SEI problem.